# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## ANSWER KEY & MARKING SCHEME SET A

Chapter 10: **Light: Shadows and Reflections** | Class 7 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** luminous | e.g. the Sun |
| 2 | **(a)** transparent | e.g. clear glass |
| 3 | **(a)** a mirror | three items needed |
| 4 | **(b)** reflection | mirror |
| 5 | **(c)** lateral inversion | e.g. AMBULANCE |
| 6 | **(d)** blurred | overlap of rays |
| 7 | **(c)** an angle to each other | creates symmetry |
| 8 | **(c)** diffused reflection | scattering |
| 9 | **(d)** d-like (laterally inverted) | left-right swap |
| 10 | **(b)** travels in straight lines | straight-line travel |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Luminous give own light, e.g. Sun (1); non-luminous only reflect light, e.g. Moon (1). | 1+1 |
| 12 | Transparent let light pass fully (1); translucent partly; opaque block it (1). | 1+1 |
| 13 | Light source, opaque object, screen (2). | 2 |
| 14 | Light travels in straight lines and is blocked (2). | 2 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Use three cards with holes in a line (1); light seen only when holes aligned (1); bend one card and light stops (1). | 1+1+1 |
| 16 | Closer to the source makes a larger shadow (1.5); farther makes a smaller shadow (1.5). | 1½+1½ |
| 17 | Light travels straight (1); opaque object blocks it (1); dark region on screen (1). | 1+1+1 |
| | **OR:** They let some light pass (1.5); opaque ones block all (1.5). | 1½+1½ |
| 18 | Virtual, erect (1); same size (1); laterally inverted, equal distance behind (1). | 1+1+1 |
| | **OR:** Box with a small hole on one side (1); translucent screen at the other (1); inverted image forms (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) Own light vs reflected (2). (ii) Sun, bulb; Moon, book (2). (iii) Reflects sunlight (1). | 2+2+1 |
| | **OR:** (i) Ball blocks light (1). (ii) Shadow grows larger (1). (iii) Lighter shadow (1). (iv) Straight-line travel (2). | 1+1+1+2 |
| 20 | (i) Transparent, translucent, opaque (3). (ii) Opaque (1). (iii) Block all light (1). | 3+1+1 |
| | **OR:** (i) Left-right reversed (1). (ii) Lateral inversion (1). (iii) 40 cm behind the mirror (1). (iv) Virtual, erect, same size (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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# ANSWER KEY & MARKING SCHEME SET B

Chapter 10: **Light: Shadows and Reflections** | Class 7 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** Sun | luminous |
| 2 | **(d)** translucent | partly allows light |
| 3 | **(a)** opaque objects | blocks more light |
| 4 | **(c)** virtual and erect | cannot be on a screen |
| 5 | **(a)** reads correct in a rear-view mirror | drivers see correctly |
| 6 | **(c)** two plane mirrors | see over obstacles |
| 7 | **(a)** vacuum (space) | light needs no medium |
| 8 | **(c)** reflection | law of reflection |
| 9 | **(c)** flat reflecting surface | e.g. a dressing mirror |
| 10 | **(c)** opaque | blocks sunlight |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Angle of incidence equals angle of reflection (2). | 2 |
| 12 | Left-right reversal of an image in a plane mirror (2). | 2 |
| 13 | It reads correctly in the rear-view mirror of vehicles ahead (2). | 2 |
| 14 | Virtual and erect (1); same size, equal distance behind mirror (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Two parallel mirrors at 45° (1.5); light reflects twice to the eye (1.5). | 1½+1½ |
| 16 | Three mirror strips in a tube (1.5); beads reflected repeatedly create patterns (1.5). | 1½+1½ |
| 17 | Mirror is smooth and gives regular reflection (1.5); wall is rough and diffuses light (1.5). | 1½+1½ |
| | **OR:** Many rays from one point reach different places (1.5); images overlap (1.5). | 1½+1½ |
| 18 | Shadow: dark patch from blocked light (1.5); image: inverted copy of the object's shape and colours (1.5). | 1½+1½ |
| | **OR:** Image is 30 cm behind the mirror (1.5); total distance 60 cm (1.5). | 1½+1½ |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) Box, hole, screen (2). (ii) Inverted (1). (iii) Larger hole gives blurred, brighter image (2). | 2+1+2 |
| | **OR:** (i) Two reflections (2). (ii) Submarines, over crowds (2). (iii) Parallel at 45° (1). | 2+2+1 |
| 20 | (i) Patterns (1). (ii) Multiple reflections (1). (iii) Inclined to each other (1). (iv) Periscope (2). | 1+1+1+2 |
| | **OR:** (i) i = r (1). (ii) Beam on mirror, measure (2). (iii) From smooth surface (2). | 1+2+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.